

Legend — reading the pinouts notation

/NAME Active LOW — true / asserted when the pin is at logic **LOW** (~0 V). A leading / (datasheets often use an overline) marks these: e.g. /RES /IRQ /CS /WE.

NAME Active HIGH / normal — asserted when the pin is **HIGH** (~+5 V). No slash. R/W is a level: HIGH = read, LOW = write.

Colour key: **Vcc** / **Vdd** = **power** · **GND** / **Vss** = **ground** · ϕ = **clock**. **Orientation:** chip seen from the top, notch up — pin 1 top-left, numbers run **down the left** then **up the right**.

Z80A CPU 40-DIP

A11	1	40	A10
A12	2	39	A9
A13	3	38	A8
A14	4	37	A7
A15	5	36	A6
CLK	6	35	A5
D4	7	34	A4
D3	8	33	A3
D5	9	32	A2
D6	10	31	A1
Vcc	11	30	A0
D2	12	29	GND
D7	13	28	/RFSH
D0	14	27	/M1
D1	15	26	/RESET
/INT	16	25	/BUSRQ
/NMI	17	24	/WAIT
/HALT	18	23	/BUSAK
/MREQ	19	22	/WR
/IORQ	20	21	/RD

ColecoVision CPU @ 3.58 MHz. **Fails:** dead console, no boot screen.

TMS9928A VDP video 40-DIP

/RAS	1	40	XTAL2
/CAS	2	39	XTAL1
AD7	3	38	R-Y
AD6	4	37	GROMCLK
AD5	5	36	Y
AD4	6	35	B-Y
AD3	7	34	/RESET
AD2	8	33	Vcc
AD1	9	32	RD0
AD0	10	31	RD1
R/W	11	30	RD2
Vss	12	29	RD3
MODE	13	28	RD4
/CSW	14	27	RD5
/CSR	15	26	RD6
/INT	16	25	RD7
CD7	17	24	CD0
CD6	18	23	CD1
CD5	19	22	CD2
CD4	20	21	CD3

Video display processor with its own 16KB VRAM. Drives the picture through an RF/video encoder. **9918A** gives COMVID; the console may use the **9928A** (Y / R-Y / B-Y). **Fails:** no / garbled picture, missing sprites or colour.

SN76489AN PSG sound 16-DIP

D5	1	16	Vcc
D6	2	15	D4
D7	3	14	CLOCK
/READY	4	13	D3
/WE	5	12	D2
/CE	6	11	D1
AUDIO OUT	7	10	D0
GND	8	9	NC

3 square-wave tones + 1 noise channel. **Fails:** no / distorted sound, dead channel.

Memory RAM

CPU RAM 1KB static work RAM

VRAM 16KB — 8× 4116 (±12/5/-5V)

BIOS 8KB OS-7 ROM

Fails: won't boot (CPU RAM) / corrupt graphics (VRAM). The 4116s need all three rails.

Test & Repair tips

No power-up Z80 / clock / BIOS ROM

No picture VDP or its VRAM

Bad graphics VRAM (4116) — check -12V

No sound SN76489

Controllers dead decoders / VDP scanline IRQ
Cartridge slot contacts are a common fault — clean before chip-swapping. Socket the VDP and PSG to test.